Pre-submission inquiry to Genome Medicine

Date: 29 October 2026

Subject: Pre-submission inquiry: unit-aware analysis of single-cell kidney disease states

Dear Editors,

We are considering submitting our manuscript, "Unit-aware analysis reveals opposing donor and within-donor disease signals in human kidney single-cell data", as a Research Article to Genome Medicine.

Single-cell disease studies often describe a cell state as disease-associated without specifying whether the signal reflects donor composition, local neighborhood structure, or coupling within a donor. We developed a unit-aware framework and evaluated it in 106,851 human kidney thick ascending limb cells from 223 donors. In aTAL2, donor-level and neighborhood-expression directions were positive while the within-donor association was negative. A prespecified 400-gene panel showed donor-within reversal in 19.0% of genes versus 12.7% of expression-matched random sets. A conditional effect-pairing null gave 20.8%, showing that the initial excess reflected the selected effect-size distribution rather than broad negative coupling between units. A smaller neighborhood-versus-within excess remained threshold sensitive. A simulation benchmark distinguished true opposing effects from concordant and null scenarios.

Two non-KPMP cohorts supported donor-level PAPPA2 disease elevation, although exact subtype replication and the external within-donor effect were not obtained. Kidney tubulointerstitial eQTL, PTWAS, and snATAC analyses are reported as regulatory context and boundary evidence rather than causal proof. The analysis code and derived data will be archived with a new persistent DOI before formal submission.

We believe this work fits Genome Medicine because it addresses a general measurement problem in genomic medicine and provides a reusable framework for cell-state studies across diseases.

The authors have no publication funding. Before submitting, may we ask whether the manuscript would be considered within scope and whether a discretionary APC waiver could be considered if the manuscript is sent for review? Genome Medicine currently lists an APC of GBP 3,690 / USD 5,490 / EUR 4,490.

We would be grateful for your guidance.

Yours sincerely,

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